AI-Enabled Digital Twin for Intelligent Building Monitoring and Predictive Analytics Using IoT

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Abstract*— Constant monitoring of environmental conditions and occupant activities in various spaces is essential for effective building management. Traditional monitoring systems can collect sensor measurements but have limitations in organizing the data for analysis and visualization in a timely manner. This paper presents an AI-enabled Digital Twin system that integrates the Internet of Things (IoT), machine learning, and rule-based analysis with real-time data processing and Digital Twin visualization for intelligent building monitoring. The system is built on an Espressif Systems ESP32 microcontroller and is equipped with DHT22 and MQ-2 sensors, as well as PIR motion sensors, to capture environmental conditions in three spaces of the prototype building: Reception, Manager Cabin, and Workspace. The measured data are transmitted to a FastAPI backend via Wi-Fi for storage in a MongoDB database. For analysis, the data are preprocessed, and a machine learning model based on Random Forest is developed to predict room occupancy, while rule-based analysis classifies environmental conditions into normal, warning, and critical states. PIR sensors also provide occupancy-related information. A Digital Twin dashboard is developed using React and provides occupancy-related information along with sensor measurements, historical data, alerts, and condition status of the monitored spaces. The integrated prototype establishes a basis for building automation and monitoring applications and demonstrates the integration of various technologies, including IoT, AI/ML, and Digital Twin.

Keywords—Digital Twin, Internet of Things, Smart Building, Machine Learning, Random Forest, ESP32, Real-Time Monitoring, Occupancy Prediction, Anomaly Detection, Building Monitoring.

I. INTRODUCTION

Building consist of multiple spaces that may vary in conditions throughout the day. Tracking factors such as temperature and humidity of a space may help understand activities in the space. There are numerous ways of sensing and capturing information. The Internet of Things (IoT) focuses on distributed sensing. Once sensors capture data, methods to help analyze the data are critical. Digital Twin (DT) helps bridge the gap of capturing data and representing it in a manner that is analogous to the real world. The representation helps break down, analyze, and monitor conditions of a system [1], [2]. Recent developments focused on bridging IoT and Machine learning in a enabling real time environment monitoring in buildings [3]. While progress is being made in the field, it is limited to certain spaces. Progress is made in other areas for real time building environment monitoring. Since numerous spaces in a building may require different types of sensors, methods to combine, manage and represent the captured data in a coherent format is essential for building environment monitoring.

AI and ML can analyze building data to recognize patterns and assist in condition evaluation. Several research studies have focused on AI and Digital Twins to optimize energy use in buildings and develop smart buildings [4]. Other research studies have used ML within Digital Twin frameworks to estimate indoor temperatures [5]. Other research focused on Digital Twins for real time building data visualization and building commissioning [6]. Still other research has used Digital Twins to analyze building occupants' thermal comfort and performance [7], [8]. Predicting indoor environmental quality (IEQ) of building spaces using AI and ML has also been extensively researched [9]. Integrating room-level data sensors with ML and deep learning techniques to process data and analyze environmental conditions and combining these with Digital Twin, internet of things (IoT) and other associated technologies can facilitate analysis of a broad spectrum of buildings.

An AI-based Digital Twin system is developed to monitor a prototype three-room building (Reception, Manager Cabin, Workspace) in this study. Real time measurements of surrounding environment in the prototype are acquired using an ESP32 and a array of sensors (DHT22, MQ-2 and PIR). Measurements are transmitted to a FastAPI server through Wi-Fi. Server stores the data in MongoDB. Data are analyzed after preprocessing. Room occupation prediction is performed using a machine learning model based on Random Forest [10]. Condition of environment in the monitored space is determined using rules. Based on rules, condition of environment is classified to normal, warning and critical state. PIR sensor is used to determine occupation of space. Digital Twin of

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monitored space developed using React. Occupation and condition of monitored space is determined using rules and sensors respectively. System acquires, analyses and communicates condition of monitored space using AI and ML, and alerts user using physical indicators. The importance of intelligent building monitoring is also reflected in the significant energy consumption of buildings, which account for about 30% of global final energy consumption and more than half of global electricity consumption , as shown in Fig. 1.

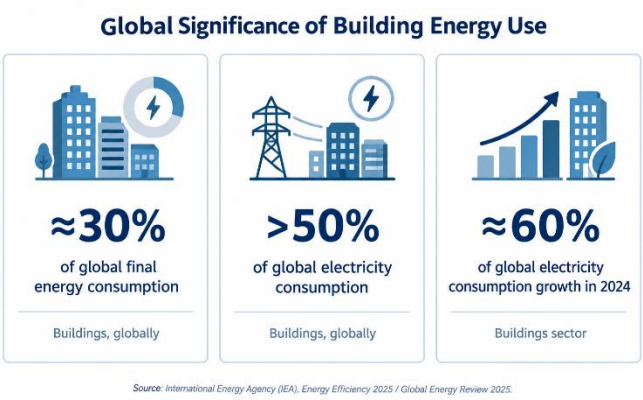


Fig. 1. Global significance of building energy use.

II. Related Works

Research on Digital Twin and IoT technology for smart buildings monitoring and analyzing has been performed recently. Boukaf et al. gave a review on Digital Twin technology for building energy consumption forecasting [11]. In an effort to improve smart buildings, Hidayat et al. developed a Digital Twin system that combines blockchain and mixed reality (MR) [12]. In the field of district heating, Tuutijärvi et al. have conducted a study on using AI in DR [13]. Fang has had experiences on Artificial Intelligence and Big Data for Smart Buildings [14]. The studies employ Digital Twins in conjunction with AI and IoT technologies, data and analysis to support the monitoring and operational decisions of buildings.

Further studies have integrated AI, IoT and Digital Twin technologies for smart and sustainable building environments [15]. Other research covered Digital Twins for smart spaces beyond traditional IoT analytics [16]. Virtual Digital Twins of office spaces have also been designed using computer vision for room occupancy and energy estimation [17]. Building Digital Twins have been developed using a combination of Building Information Modeling and Computer Vision [18]. Scalable Digital Twin architectures at the edge have also been examined [19]. Web-based Building Information Modeling (BIM) platforms have been designed for building Digital Twins [20]. However, these studies generally address specific aspects such as energy analysis, visualization, occupancy estimation, BIM, computer vision, or scalable architectures. The proposed system combines IoT-based environmental sensing, Random Forest-based room occupancy prediction, rule-based environmental condition analysis, real-time backend processing, physical alerts, and room-level Digital Twin visualization within a single prototype building-monitoring system. A comparison of related systems is presented in Table I.

TABLE I. Comparison of Related Digital Twin-Based Building Monitoring Systems

Article Title	Key Objective	Technologies Used	Limitation
AI-Powered Digital Twins and Internet of Things for Smart Cities and Sustainable Building Environment	Discusses the application of AI, Digital Twins and IoT in smart city and sustainable building.	Artificial Intelligence, Digital Twin, IoT, smart-building technologies, and sustainability-oriented analysis.	Provides a broad technology perspective rather than implementing a room-level monitoring prototype with physical alerts.
Digital Twins for Smart Spaces—Beyond IoT Analytics	Explores how Digital Twins can extend smart-space management beyond conventional IoT-based data analysis.	Digital Twin concepts, IoT sensing, smart-space data, and physical–virtual system integration.	Focuses on broader smart-space management and does not implement the complete room-level monitoring and alert mechanism used in this work.
VR digital twin of office space with computer vision-based estimation of room occupancy and power consumption	Develops a virtual representation of an office space for estimating occupancy and power consumption.	Virtual Reality, Digital Twin, computer vision, environmental sensing, occupancy estimation, and power-consumption analysis.	Concentrates on computer-vision-based occupancy and power estimation rather than combining multiple environmental sensors with rule-based condition analysis and physical alerts.
Computer Vision Enabled Building Digital Twin Using Building Information Model	Creates a Digital Twin of a building using BIM and computer vision.	BIM, computer vision, building data, spatial representation, and Digital Twin technologies.	Emphasizes BIM and vision-based building representation rather than multi-parameter IoT sensing and room-level environmental condition monitoring.
Web-Based BIM Platform for Building Digital Twin	Builds a web-based BIM platform for representing and managing building Digital Twin information.	BIM, web-based visualization, building information, and Digital Twin representation.	Focuses mainly on BIM-based Digital Twin representation and does not integrate machine-learning-based occupancy prediction with multi-sensor condition analysis.

III. Methodology

The proposed system integrates IoT sensing, data communication, machine learning, rule-based condition analysis, and Digital Twin visualization for real-time building monitoring. Sensor data are collected from the physical prototype using ESP32-based sensing units and transmitted to the backend through Wi-Fi. The received data are stored, preprocessed, analysed, and presented through a room-level Digital Twin dashboard. A Random Forest machine learning model is trained for room occupancy prediction, while rule-based analysis is used to identify abnormal environmental conditions and generate corresponding status levels.

A. System Architecture

The proposed system is designed to have five layers: hardware layer, communication layer, backend layer, AI and analytics layer, and application layer. The hardware layer is the ESP32 and the sensors DHT22, MQ-2 and PIR that are placed in the prototype rooms. The ESP32 reads the sensor data and communicates with Wi-Fi to the FastAPI backend. Data received is stored in a Mongo database, ready for pre-processing and analysis.The AI and analytics layer includes the data preprocessing, Random Forest based room occupancy model, and rule based anomaly analysis. The machine learning model predicts the room occupancy based on occupancy-related features of the sensors and the rule-based model is used to deduce the normal, warning or critical conditions based on environmental parameters. This processed information can then be visualised on the React-based Digital Twin dashboard, creating a bridge between the physical building and the virtual one. The proposed system is depicted in the entire process in Fig. 2

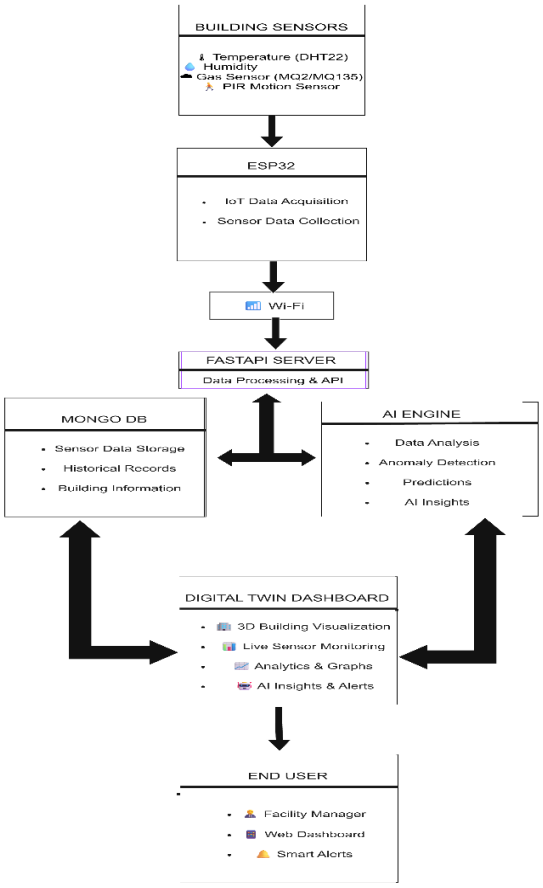


Fig. 2. Overall workflow for AI-enabled Digital Twin system.

The five-layer organization separates sensing, communication, backend processing, AI and analytics, and application-level visualization. This organization allows sensor acquisition, data processing, machine learning analysis, condition assessment, and Digital Twin visualization to operate as connected stages. The layer-wise organization of the proposed system is presented in Fig. 3.

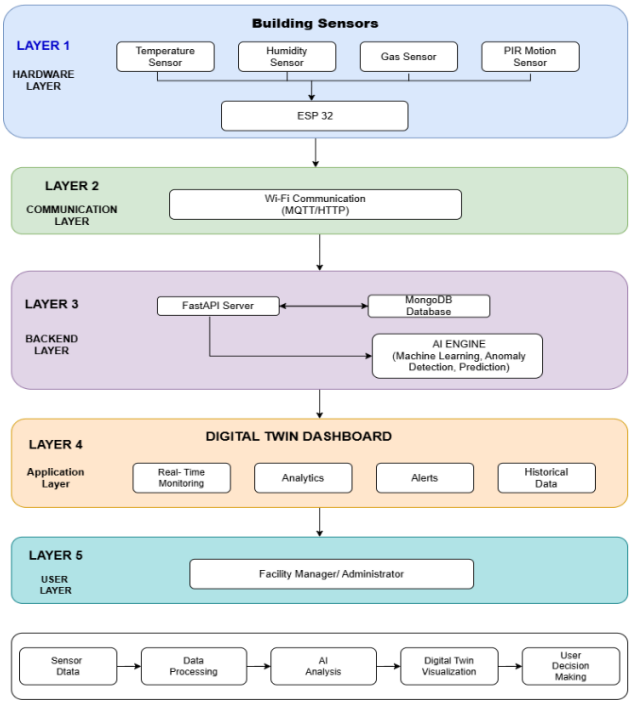


Fig. 3. Proposed five-layer architecture for AI-based Digital Twin building monitoring.

B. Sensor Data Acquisition

Physical prototype is comprised of three monitored spaces: Reception, Manager Cabin, and Workspace. The main controller used to collect data from the connected sensors is an ESP32. The DHT22 sensor is used to get the temperature and humidity measurements, and the MQ-2 sensor is used to do gas monitoring. The PIR sensors are used for motion detection and give occupancy information. The ESP32 periodically reads the sensor values and formats them for sending to the backend.The real-time building data acquisition system based on ESP32 sensing is depicted as shown in Fig. 4.



Fig. 4. ESP32-based sensor setup used for building data acquisition.

C. Data Transmission and Storage

ESP32s use Wi-Fi to send the gathered sensor data to the FastAPI backend. The readings are transferred to the backend where they are linked with the room and sensor parameters. The data received is stored in MongoDB, where the current measurements and historical readings can then be accessed by the application. This is a steady stream of data between the physical prototype and the Digital Twin environment.The data transmission and storage process implemented in the system is shown in Fig. 5.

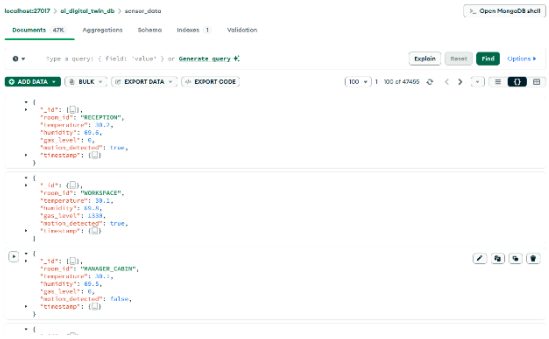


Fig. 5. Data transmission and storage process of the proposed system.

D. Data Preprocessing and AI/ML Analysis

The collected sensor data is preprocessed by extracting temperature, humidity, gas level, and motion values and normalizing them for analysis. A Random Forest classifier is trained using environmental and PIR sensor features to predict room occupancy.In parallel, a rule-based anomaly detection method classifies building conditions as normal, warning, or critical based on predefined thresholds and generates a risk score. The processed results are used by the Digital Twin dashboard for monitoring and alerts. As shown in Fig. 6.

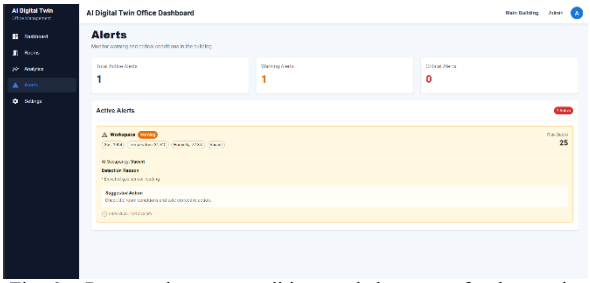


Fig. 6. Processed sensor conditions and alert states for the monitored rooms.

E. Digital Twin Dashboard

The processed information is represented in the Digital Twin dashboard as room-level information for Reception, Manager Cabin, and Workspace**.** The dashboard displays current sensor values, occupancy-related information, historical trends, status indicators, and warning conditions. This visualization allows the user to observe the condition of individual spaces through a unified interface.Physical alert mechanisms, including the buzzer and the Manager Cabin LED connected to the PIR detection system, provide local indications in addition to the software dashboard. The developed Digital Twin dashboard for real-time building monitoring is shown in Fig. 7.

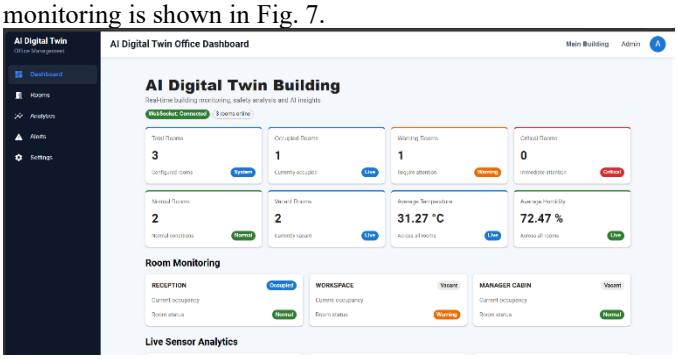


Fig. 7. Data visualization for building that is provided for monitoring in real time.

IV. Experimental Setup and Results

The system was implemented and tested using a three-room building prototype consisting of Reception, Manager Cabin, and Workspace. The setup integrates ESP32-based sensors, Wi-Fi, FastAPI, MongoDB, AI-based analysis, and a React Digital Twin dashboard. The experiments verified sensor acquisition, data transmission, storage, monitoring, abnormal-condition detection, and alerts.

A. Hardware and Software Implementation

In the prototype the ESP32 was attached with DHT22 sensor which is for measurement of temperature and humidity, meanwhile with MQ-2 sensor which is for measuring the gas and with PIR sensor which is for measuring the presence of human. It uses sensor data to send it to FastAPI via Wi-Fi to store it in MongoDB. Processed data is presented on the React dashboard and sent to a buzzer and Manager Cabin LED provide local alert. The hardware configuration is depicted in fig 8.



Fig. 8. Experimental hardware setup with ESP32 and connected building sensors.

B. Real-Time Sensor Monitoring

The ESP32 successfully collected temperature, humidity, gas, and motion data from all three rooms. The readings were transmitted to the backend, stored in MongoDB, and displayed on the Digital Twin dashboard for room-level monitoring. The real-time sensor values displayed by the developed system are shown in Fig. 9.

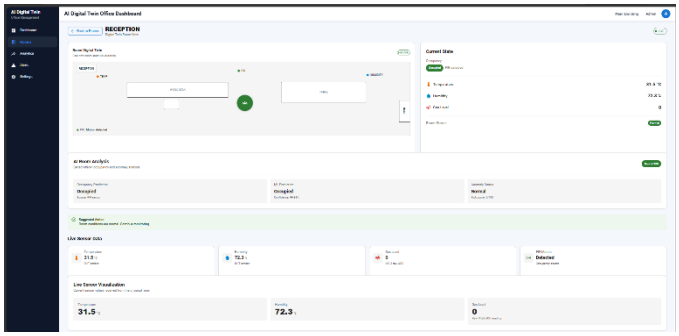


Fig. 9. Real-time room-level sensor monitoring through the Digital Twin dashboard.

C. Abnormal Condition and Alert Detection

The system was tested under different sensor conditions using threshold-based classification to identify normal, warning, and critical states. For a measured parameter, the status is determined as

$$S(x) = \begin{cases} \text{Normal}, & x < T_1 \\ \text{Warning}, & T_1 \le x < T_2 \\ \text{Critical}, & x \ge T_2 \end{cases} \qquad (1)$$

where T_1 and T_2 represent the defined warning and critical thresholds, respectively.

PIR-based occupancy-related detection was represented as

$$O = \begin{cases} 1, & \text{if motion is detected} \\ 0, & \text{otherwise} \end{cases} \qquad (2)$$

where 1 indicates detected motion and 0 indicates no detected motion. Abnormal conditions were displayed on the dashboard, while the buzzer and Manager Cabin LED provided physical alerts. The warning and alert conditions observed during testing are shown in Fig. 10.

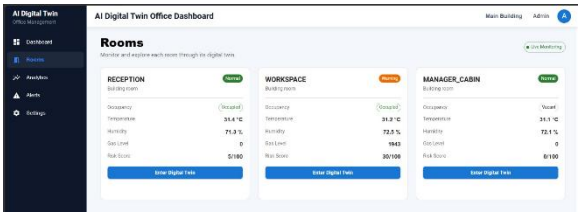


Fig. 10. Room-level Digital Twin dashboard showing environmental and occupancy conditions.

D. Data Analysis and Historical Observation

MongoDB is used to store sensor data over time. We can then go back and see historical measurements of various conditions like temperature, humidity, gas levels, and motion. We can then view these conditions in the Digital Twin dashboard to provide a room level view. By having a history of measured conditions, we can start looking for environmental condition changes and provide a basis for understanding the conditions in the monitored space. The mean of the specified sensor condition is a sum of all the sensor readings divided by the number of sensor readings sampled (as represented in Equation 1).

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad (3)$$

where (X_{\mathrm{avg}}) is the average of the readings, (X_1,X_2,\ldots,X_N) are the each sensor's reading and (N) is the number of readings. Calculated conditions and historical data are used to analyse environmental conditions in Reception, Manager Cabin and Workspace.The dashboard provides a condition comparison of monitored spaces, as well as a historical view of previously collected data for further analysis. The processed data are used to display room conditions, status indicators, and historical trends through the Digital Twin dashboard, as shown in Fig. 11.

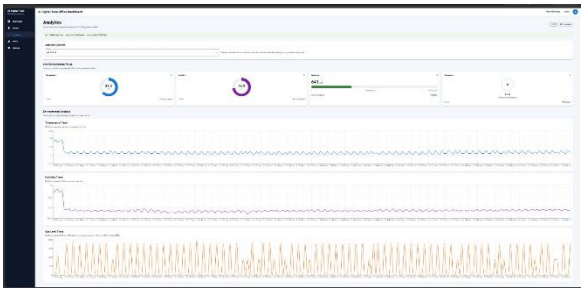


Fig. 11. Historical sensor trends and analytical information displayed in the Digital Twin dashboard.

E. Digital Twin Visualization and Results

F.

The three rooms being monitored are represented virtually on the Digital Twin dashboard which shows the values of the sensors, occupancy-related status, historical data and indicators of warning conditions. During testing, changes in the physical sensor conditions were reflected in the corresponding digital room status. The local alerts were available through the buzzer and the Manager Cabin LED was supported by the system.

G. Experimental Observations

The experiments verified the complete data flow from physical sensing to Digital Twin visualization. ESP32

successfully acquired and transmitted sensor data, FastAPI processed the readings, MongoDB stored the data, and the dashboard displayed room-level information. Threshold testing verified the warning and critical status indications, while PIR testing confirmed motion detection and Manager Cabin LED operation. These results demonstrate the practical integration of IoT sensing, backend processing, AI-based analysis, alert mechanisms, and Digital Twin visualization in the proposed building-monitoring system. The key implementation challenges observed during development and the measures used to address them are presented in Table II.

TABLE II. IMPLEMENTATION CHALLENGES AND CORRESPONDING SOLUTIONS

Challenge Encountered	Solution Implemented
Unstable sensor readings during initial hardware testing	Sensor connections and power supply were checked, and the sensor readings were repeatedly tested to ensure stable data acquisition.
Difficulty in maintaining reliable communication between ESP32 and backend	Wi-Fi communication was configured and the ESP32 data transmission process was tested with the FastAPI backend.
Mapping sensor readings to the correct prototype rooms	Room identifiers were associated with the transmitted sensor data so that each reading could be displayed under its corresponding room.
Handling different sensor parameters in a unified system	Temperature, humidity, gas, and motion readings were structured into a common data format before backend processing and storage.
Identifying abnormal environmental conditions	Threshold-based conditions were implemented to classify readings into normal, warning, and critical states.
Synchronizing physical sensor conditions with the Digital Twin dashboard	The backend continuously processed the received sensor data and provided updated information to the React-based dashboard.
Providing physical indication for detected motion and alerts	A buzzer was integrated for alert indication, and an LED was connected to the Manager Cabin PIR sensor for motion-based indication.

V. CONCLUSION

The developed system demonstrates an AI-enabled Digital Twin for intelligent building monitoring by integrating IoT sensing, AI-based analysis, backend services, and a room-level Digital Twin dashboard. The ESP32 successfully collected temperature, humidity, gas, and motion data from the Reception, Manager Cabin, and Workspace and transmitted the readings to the FastAPI backend for MongoDB storage. The React dashboard provided real-time monitoring, historical observation, occupancy-related status, and identification of warning and critical conditions.The integration of threshold-based analysis, Random Forest-based occupancy prediction, and physical alerts through the buzzer and Manager Cabin LED connected digital monitoring with physical responses. Experimental testing verified the practical operation of the proposed prototype. The system can be further extended with additional sensors, advanced predictive models, energy analysis, predictive maintenance, and larger building deployments.

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